

BENJAMIN HOFFMAN

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EDUCATION

ETH ZURICH PhD Student in the Networked Systems Group Netflix Research Fellow Affiliated PhD at the ETH AI Center ▸ Current research: control, algorithms, and machine learning for large scale, complex systems and networks ▸ Supervised by Prof. Laurent Vanbever (ETH) and Prof. Gioele Zardini (MIT)	Zurich, Switzerland Apr. 2025 - present
ETH ZURICH MSc in Robotics, Systems and Control — GPA: 5.7/6.0 ▸ Focus on control and machine learning	Zurich, Switzerland Sep. 2022 - Dec. 2024
LUND UNIVERSITY Academic Exchange Semester — GPA: 5.7/6.0 ▸ Focus on control and machine learning	Lund, Sweden Jan. 2022 - Jun. 2022
ETH ZURICH BSc in Mechanical Engineering — GPA: 5.2/6.0 ▸ Focus on mechatronics and control	Zurich, Switzerland Sep. 2018 - Aug. 2021
KANTONSSCHULE WETTINGEN Swiss Matura and IB Diploma — GPA: 5.7/6.0 (top 0.5%, salutatorian) ▸ Major in law and economics, academic exchange semester at the Gymnase de Nyon (VD, Switzerland)	Wettingen, Switzerland Aug. 2014 - Jun. 2018

RESEARCH EXPERIENCE

NETFLIX Graduate Research Fellow — full time ▸ Research on classical and learning-based control algorithms for video streaming and content delivery ▸ Supervised by Dr. Te-Yuan Huang and Prof. Renata Teixeira	Los Gatos, CA, USA Sep. 2026 - present
NETWORKED SYSTEMS GROUP — ETH ZURICH Research Assistant — part time ▸ Researched and evaluated state-of-the-art classical and learning-based ABR video streaming control algorithms ▸ Supervised by Dr. Alexander Dietmüller and Prof. Laurent Vanbever	Zurich, Switzerland Jun. 2024 - Mar. 2025
MASTER THESIS — ETH AI CENTER LEARNING & ADAPTIVE SYSTEMS GROUP Sample Efficient Model-Based Reinforcement Learning — Grade: 5.75/6.0 (view here) ▸ Using model-based RL for efficient policy learning on a Boston Dynamics Spot quadruped with JAX, PyTorch, and C++ ▸ Supervised by Jin Cheng, Chenhao Li, Dr. Bhavya Sukhija, Dr. Lenart Treven, Prof. Stelian Coros, and Prof. Andreas Krause	Zurich, Switzerland Jun. 2024 - Dec. 2024
SEMESTER THESIS — NETWORKED SYSTEMS GROUP Machine Learning Based Video Streaming Algorithms — Grade: 6.0/6.0 (view here) ▸ Developed a network-trace collection and ABR algorithm benchmarking system using Docker, Python, and AWS/Azure SDKs ▸ Supervised by Dr. Alexander Dietmüller and Prof. Laurent Vanbever	Zurich, Switzerland Sep. 2023 - Jan. 2024
BACHELOR THESIS — COMPUTATIONAL ROBOTICS LAB State Estimation for Legged Robots — Grade: 6.0/6.0 (view here) ▸ Researched, developed and implemented different quadruped state-of-the-art state estimation algorithms using C++ ▸ Supervised by Dr. Dongho Kang and Prof. Stelian Coros	Zurich, Switzerland Feb. 2021 - Aug. 2021

WORK EXPERIENCE

NETFLIX Research Intern — full time intern (fellowship offer received) ▸ Research on learning-based algorithms for Netflix's video streaming, focusing on improving performance at startup ▸ Designed, trained, and evaluated large scale machine learning models to improve video streaming quality	Los Gatos, CA, USA Jun. 2026 - Sep. 2026
BAIN & COMPANY Associate Consultant Intern — full time intern (return offer received) ▸ Advised 3 industry leading firms and Fortune 500 companies on performance improvement and IT strategy projects ▸ Conducted empirical research and analyzed large, unstructured datasets to formulate strategic recommendations	Zurich, Switzerland Feb. 2024 - Apr. 2024

ERANEOS	Zurich, Switzerland
Data Science Intern — full time intern (return offer received)	Jun. 2023 - Dec. 2023
▸ Worked on 5+ projects in finance and the public sector with a focus on ML & GenAI, LLMs, and data analytics ▸ Developed custom LLM solutions and multi-agent systems using fine-tuned local LLMs, RAG, and OpenAI API	
GÜDEL GROUP	Bern, Switzerland
Robotics and Software Engineer — full time intern and part time working student (return offer received)	Aug. 2021 - May 2023
▸ Machine learning, data science and software engineering for robotics and industrial analytics ▸ Transformed 3+ quality control procedures to ML-driven processes and rewrote the software for 4+ key applications	
ETH ZURICH	Zurich, Switzerland
Teaching Assistant for Analysis III and Dimensioning I — part time	Sep. 2020 - Jan. 2023
▸ Prepared and led weekly exercise sessions for 30+ students, graded assignments and exam supervision ▸ Elected by students to lead a two-day exam preparation course with 30+ participants for Analysis III	

SELECTED PUBLICATIONS

[1] Lukas Röllin, Sushovan Das, **Benjamin Hoffman**, and Laurent Vanbever. “A New Paradigm for Modern AI Interconnect Technology”. In: *ACM SIGCOMM HotOptics* (Denver, USA, 2026). [PDF]

[2] Rufat Asadli, **Benjamin Hoffman**, Ioannis Protogeros, Laurent Vanbever. “Evaluating Agentic Configuration Repair for Computer Networks”. In: *ICML Workshop on Agents in the Wild: Safety, Security, and Beyond* (Seoul, South Korea, 2026). [PDF]

[3] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Towards Closing the Performance Gap of Learning-Based Adaptive Bitrate Control”. In: *Under Review* (2026).

[4] Ioannis Protogeros, Rufat Asadli, **Benjamin Hoffman**, and Laurent Vanbever. “Benchmarking LLM-Driven Network Configuration Repair”. In: *Preprint* (2026). [PDF]

[5] **Benjamin Hoffman**, Alexander Dietmüller, Ayush Mishra, and Laurent Vanbever. “Into the Wild: Real-World Testing for ML-Based ABR”. In: *ACM PACMI* (Seoul, South Korea, 2025). [PDF]

[6] **Benjamin Hoffman**, Jin Cheng, Chenhao Li, and Stelian Coros. “Learning More With Less: Sample-Efficient Model-Based RL for Loco-Manipulation”. In: *CoRL Workshop on Resource-Rational Robot Learning* (Seoul, South Korea, 2025). [PDF]

[7] **Benjamin Hoffman**, Alexander Dietmüller, and Laurent Vanbever. “Poster: Towards Network Model Generalization using Strategic Data Collection”. In: *ACM SIGCOMM Posters and Demos* (Coimbra, Portugal, 2025). [PDF]

EXTRACURRICULAR EXPERIENCE

UNITED NATIONS – ETH STUDENT TEAM	Zurich, Switzerland
Project Lead	Sep. 2025 - present
▸ Building a voting-data processing pipeline and a user-facing visualization interface for all 3 main UN bodies ▸ In charge of project management, leading a team of 5 software engineers, coordinating with UN stakeholders	
GLP SIGGENTHAL	Obersiggenthal, Switzerland
Board and Committee Member Member of Local Parliament	Aug. 2021 - present
▸ Board member in charge of communications and strategic planning, as well as local campaigns and initiatives ▸ Member of local parliament (since 2026) and municipal environmental advisory body (<i>since 2022</i>)	
VEREIN LEGAL HELP (NGO)	Zurich, Switzerland
Board Member and Head of IT	Sep. 2022 - Sep. 2025
▸ In charge of strategic initiatives, maintaining long-term growth, as well as all technical and software related projects ▸ Led an organization-wide (100+ users) software transition project resulting in savings of 5000+ CHF annually	

SKILLS & CERTIFICATIONS

PROGRAMMING	Python, C++, MATLAB, JavaScript
FRAMEWORKS	PyTorch, JAX, scikit-learn, Pandas, NumPy, SciPy, Docker, AWS/Azure, htsim, netUnicorn, tcpdump, Wireshark
LANGUAGES	English (Native), German (Native), Swiss German (Native), French (B2)
CERTIFICATIONS	J&S Coach Tennis & Snowboard, Certificate in Portfolio and Risk Management (University of Geneva, 2023)

INTERESTS

SPORTS	Competitive tennis player (coach and team captain), snowboarding, running, strong interest in health and fitness
OTHER INTERESTS	Public Policy, law and economics, global and local politics, nature and the environment